

Graphing linear inequalities in two variables



1

- a No b Yes

2

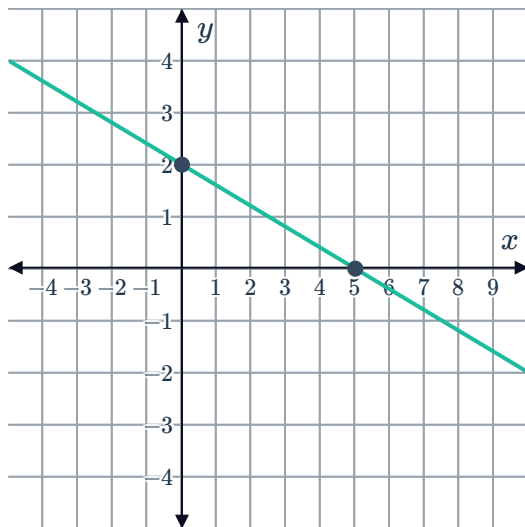
- a
i No ii No iii No iv Yes
- b
i No ii No iii No iv Yes
- c
i No ii No iii No iv Yes

3

- a $y \leq 5$ b $x \geq 2$ c $y \geq -2x - 6$ d $y < 2x + 6$
- e $y > -3x$ f $y \leq 2x + 5$ g $y \geq 3x - 3$ h $y < -2x - 4$
- i $y \leq -2$ j $y > -3x + 6$

4

- a $x = 5$ b $y = 2$
- c $2x + 5y > 10$



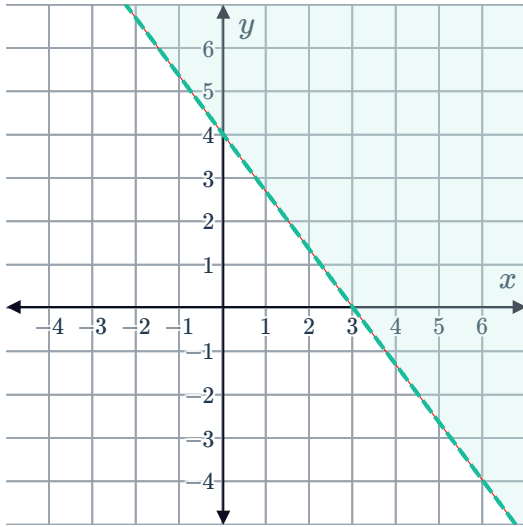
5

a $x = 3$



b $y = 4$

c



d No

6

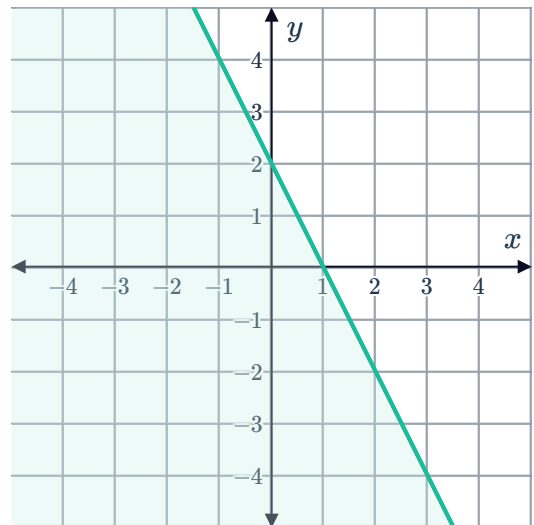
a $x = 1$

b $y = 2$

c

i No ii Yes iii No iv No

d



e Yes

7

a $x = -2$

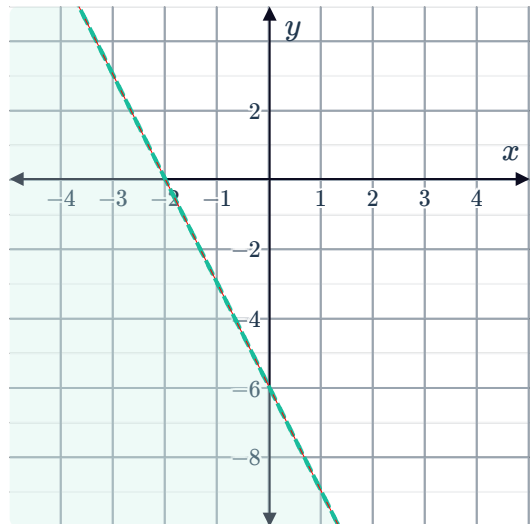
c

- i No ii No iii Yes iv No



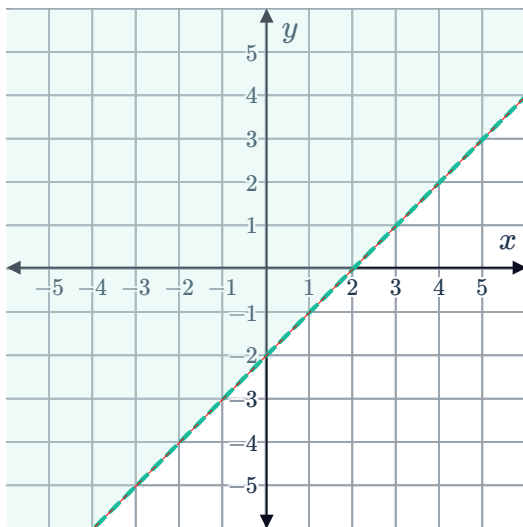
b $y = -6$

d

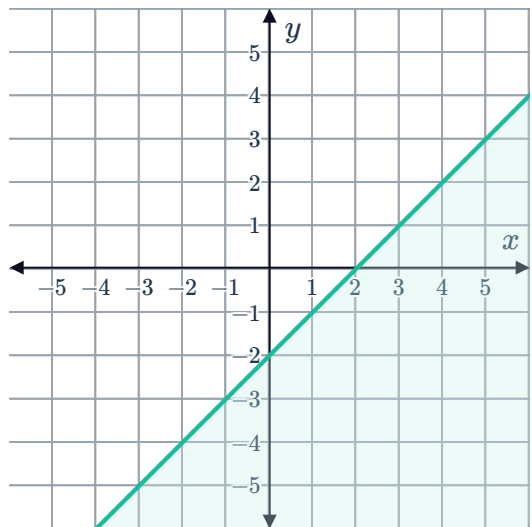


8

a

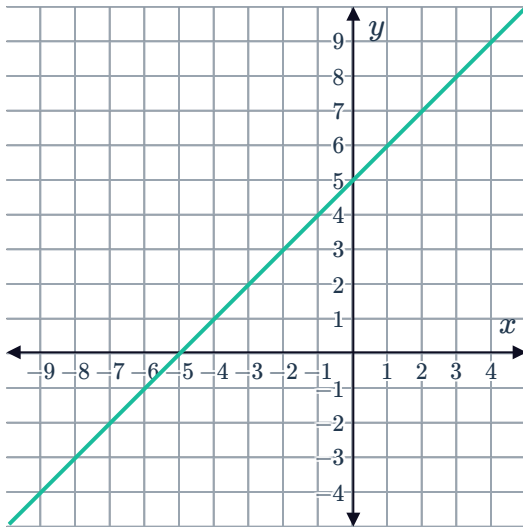


b



9

a

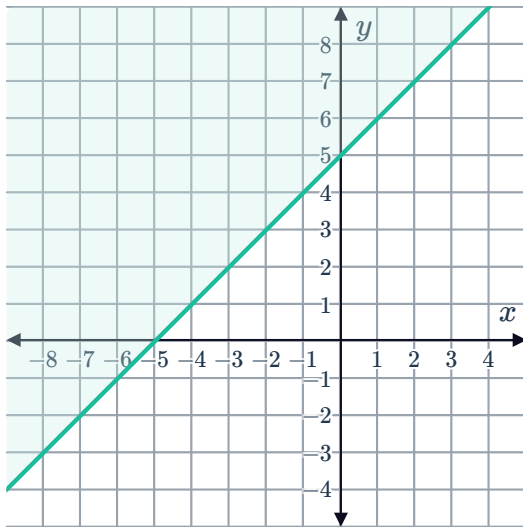


b

i Yes

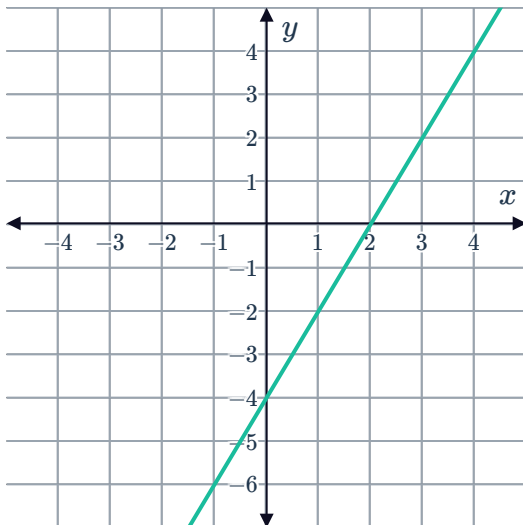
ii No

c



10

a

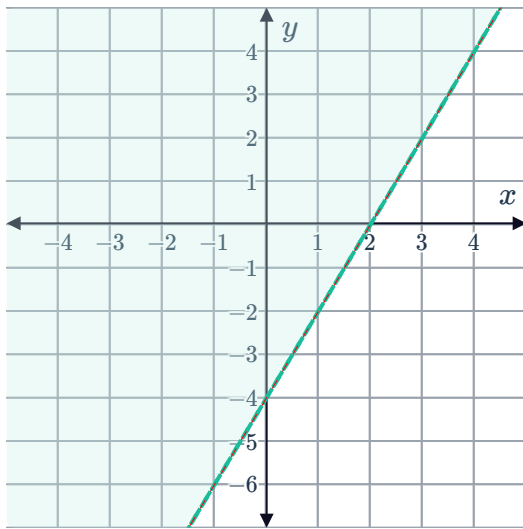


b

i No

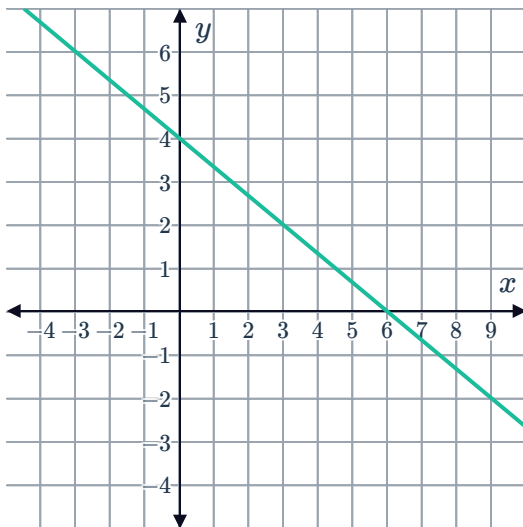
ii Yes

c



11

a

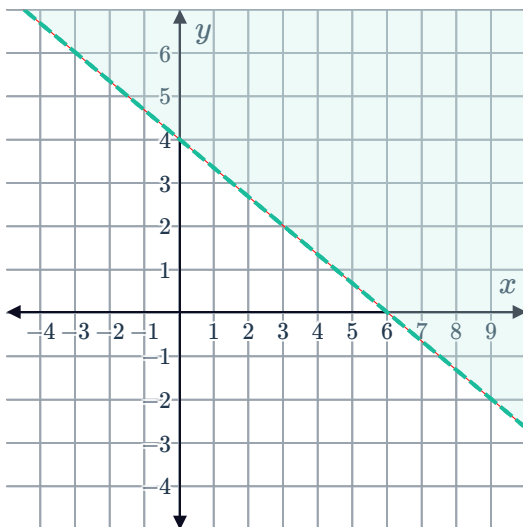


b

i No

ii Yes

c



12

a

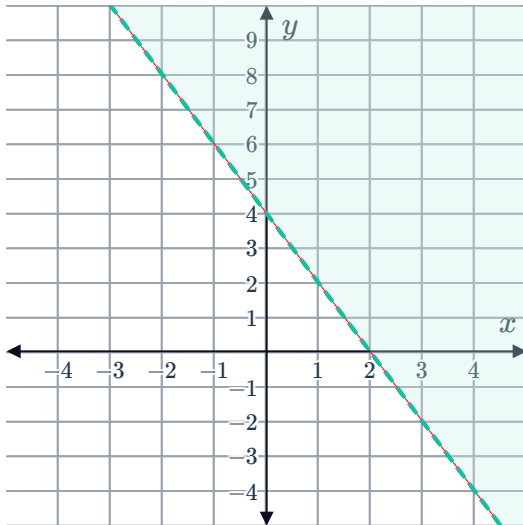
x	0	1	2
y	4	2	0



b

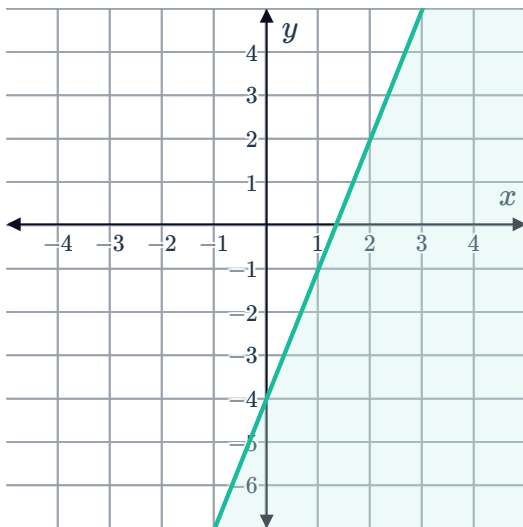
- i No ii Yes iii No iv No

c

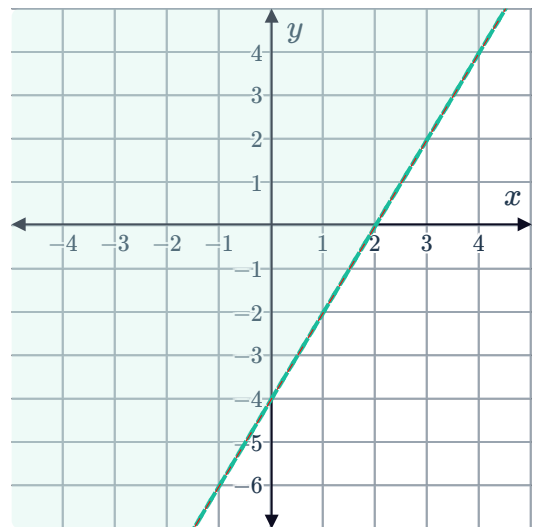


13

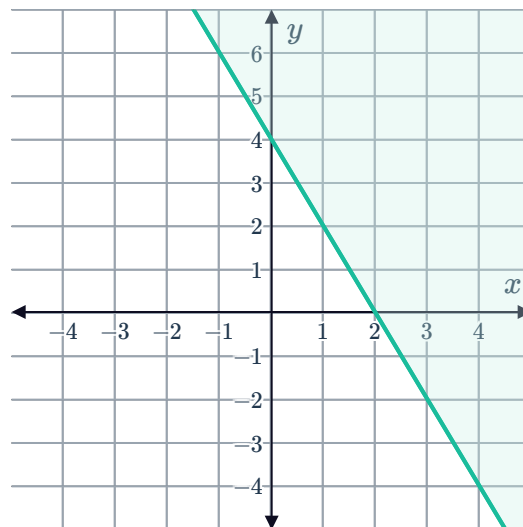
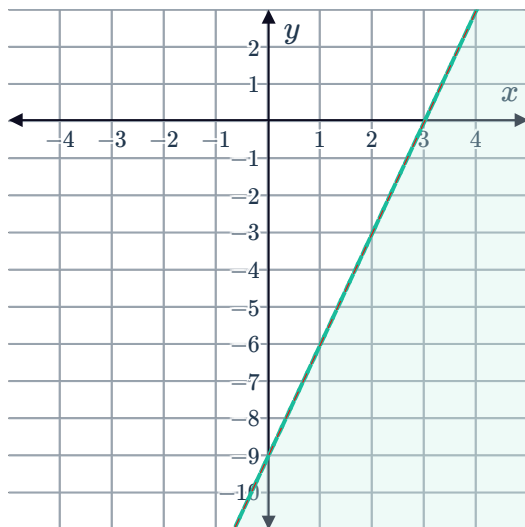
a



b



c



14

a

i 4

ii $y = 4x - 5$

iii $y \leq 4x - 5$

b

i 3

ii $y = 3x - 6$

iii $y \geq 3x - 6$

15 $y \leq 95 - \frac{19}{9}x$

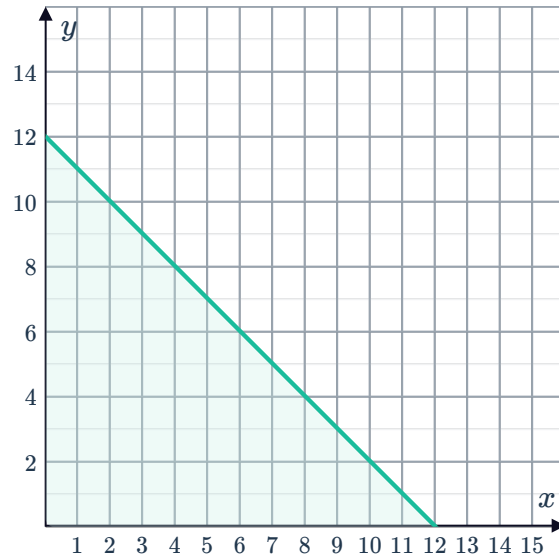
16 $y \leq 75 - \frac{15}{17}x$

17 $y \leq x - 58$

18

a $y \leq 12 - x$

b



c

i No

ii Yes

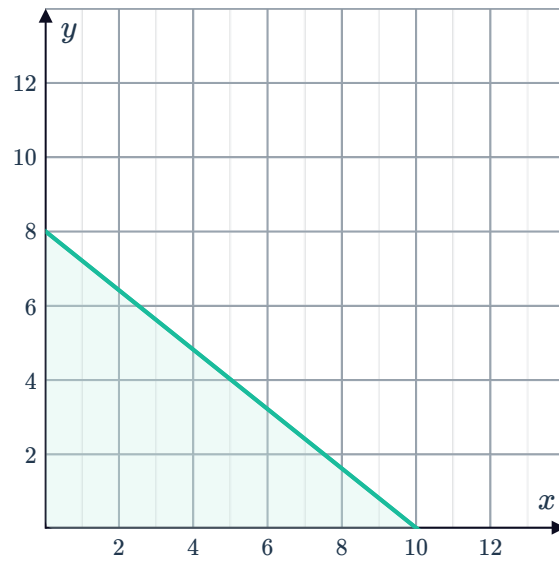
iii No

iv No

19

a $y \leq 8 - \frac{4}{5}x$

b



c

i No

ii Yes

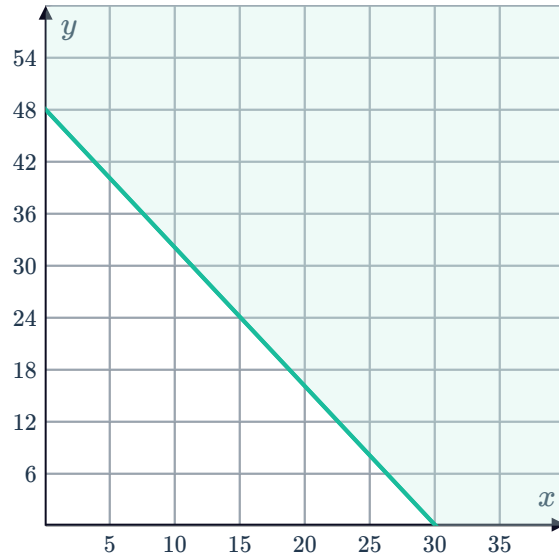
iii No

iv No

20

a $y \geq 48 - \frac{8}{5}x$

b



c

i No

ii No

iii No

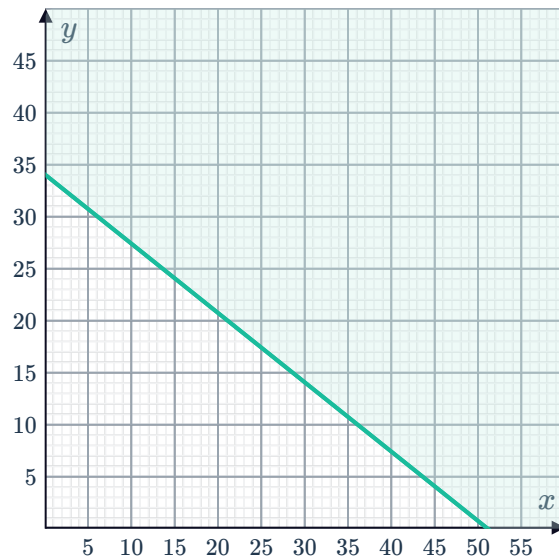
iv Yes

d 30 books

21

a $y > 34 - \frac{2}{3}x$

b

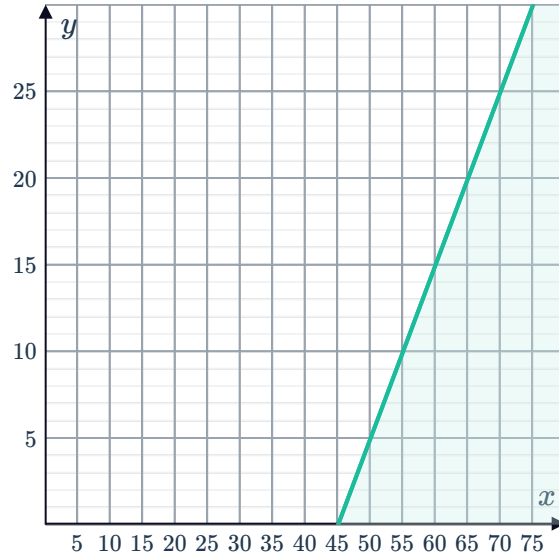


c B

22

a $y \leq x - 45$

b



c 65 minutes

Additional questions

23

a Solid

b Dashed

c Solid

d Dashed

24

a $(2, 4)$

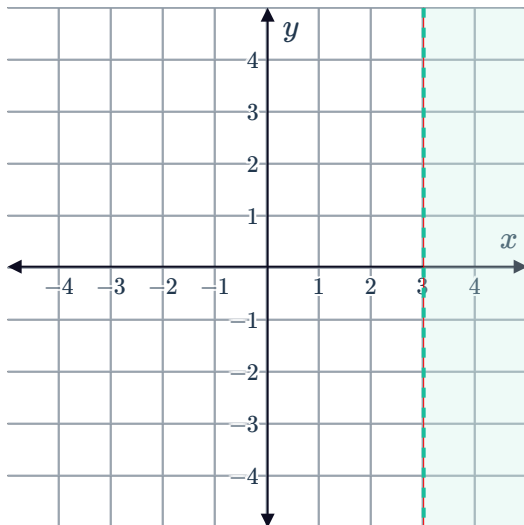
b $(2, 4)$, $(-3, 2)$ and $(-1, -6)$

25

a

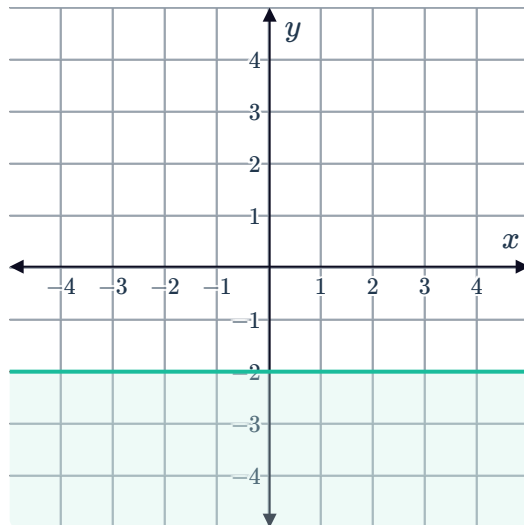
- i The point $(0,0)$ does not satisfy the inequality.

ii



- i The point $(0,0)$ does not satisfy the inequality.

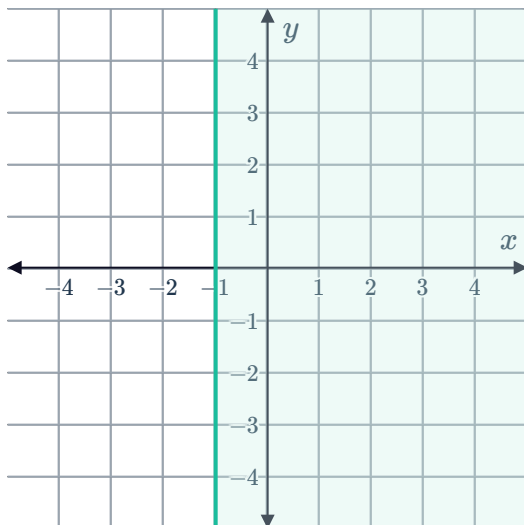
ii



c

- i The point $(,0,0)$ satisfies the inequality.

ii



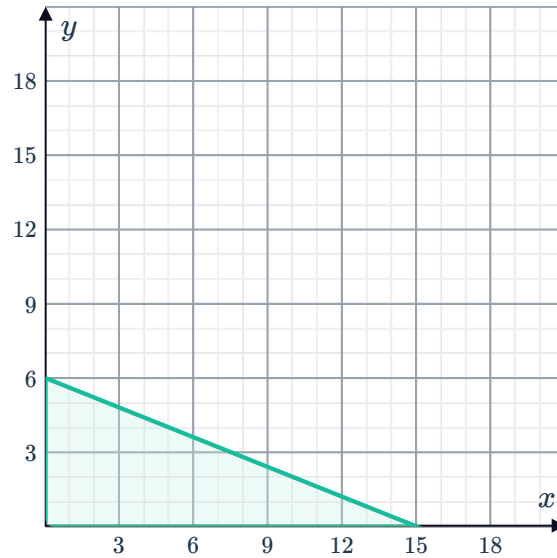
26 $x + y \geq 43$

27 $10x + 13y \geq 260$

28

a $2x + 5y \leq 30$

b

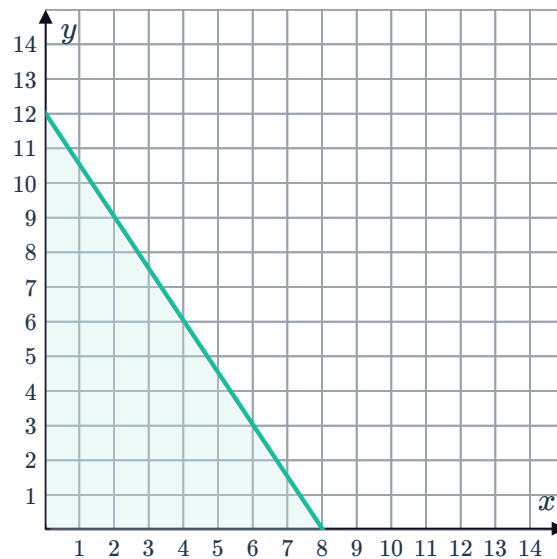


- c The x -intercept of 15 represents the greatest number of games of Table Tennis that Oprah could play if she decides not to play any games of Frog Days.
- d The y -intercept of 6 represents the greatest number of games of Frog Days that Oprah could play if she decides not to play any games of Table Tennis.

29

a $6x + 4y \leq 48$

b



- c There are 61 different ways, one for each integer coordinate inside the solution set of the inequality.